

Chat Session 3 — Component Finalisation & Feasibility Confirmation

Date: March 8, 2026
Scope: Final component buying list review, ESP32-CAM decision, jumper wire warning, project feasibility assessment

1. Session Overview

Short but decisive session. Two topics covered:

- 1. User presented their final components list before purchasing — reviewed for gaps and conflicts
- 2. User asked a direct feasibility question — full phase-by-phase assessment provided

No code was written. No architectural changes made.

2. Component Buying List — Reviewed

User's list:

- 1. ESP32-CAM MB Programming Module (Base Board)
- 2. ESP32S CAM WiFi Module with OV3660 Camera (2MP)
- 3. 400-point Solderless Breadboard (Copper)
- 4. DC Power Jack Female Connector with 2-Pin Screw Terminal PBT
- 5. MPU6050 — 6-Axis Accelerometer + Gyroscope Sensor
- 6. Jumper wires M2F + F2M — 2 sets, 10 wires each (= 20 wires total)

3. Issues Flagged

3.1 ESP32-CAM — Architecture Conflict Raised

Previous decision: User's phone (IP Webcam app) = fixed overhead camera. ESP32-CAM was previously evaluated and rejected — lower resolution, firmware setup required, ~₹400 cost vs ₹0 for phone.

New information from user: "We are buying the ESP32-CAM just in case we might replace it after some time without our mobile module."

Resolution: ESP32-CAM purchase approved as **reserve / future upgrade** — kept in a box, not integrated into current system. Current architecture (phone = overhead camera) remains unchanged.

Camera	Current Role	Status
User's phone	Fixed overhead → YOLOv8	Active — in use now
ESP32-CAM (OV3660)	Reserved for future phone replacement	Spare — stored

Decision locked: Do not replace phone with ESP32-CAM during current development phases. Revisit only if phone is unavailable or if eye-in-hand (wrist-mounted) camera is added in future scope.

3.2 Jumper Wires — Quantity Warning

Problem: 10-wire sets × 2 = only 20 wires. Minimum needed for core wiring alone:

- ESP8266 → PCA9685: 5 wires (3.3V, GND, SCL, SDA, GND-common)
- MPU6050 → I2C bus: 4 wires (VCC, GND, SCL, SDA)
- Spare for breadboard testing and debugging: ~10–15 wires

Recommendation given: Buy **40-pin Dupont sets** instead of 10-wire sets. Same price range (~₹80–120), vastly more useful.

Outcome: User noted — will buy larger sets.

3.3 Missing Items Flagged

Item	Why critical
2mm flathead screwdriver	PCA9685 screw terminal blocks (V+ / GND) physically cannot be tightened without this — critical and often forgotten
Electrical tape or heat shrink	Bare PSU wire joints from screw terminal need insulation before powering on

Confirmed NOT needed (reminder):

- Pull-up resistors → already on PCA9685 and MPU6050 boards
- Logic level converter → ESP8266 3.3V is directly compatible with PCA9685 VCC

4. Project Feasibility Assessment

User asked: **"With this current setup is the project that we are doing possible or not?"**

Answer: Yes — completely possible. Hardware is overqualified.

Phase-by-phase breakdown:

Phase	Feasibility	Reason
Phase 1 — MoveIt2	✅ Fully possible	ROS2 Jazzy + MoveIt2 official support; clean 12-link URDF ready; <code>demo.launch.py</code> already running
Phase 2 — Gesture Control	✅ Fully possible	MediaPipe verified; RTX 3060 is overkill (runs on CPU); mapping is pure math
Phase 3 — YOLOv8 Sorting	✅ Fully possible	RTX 3060 + CUDA → ~15ms inference; better than most research setups; YOLOv8 already installed

Phase	Feasibility	Reason
Phase 4 — MQTT Remote	✅ Fully possible	HiveMQ free = 100 connections, no time limit; ngrok works for demos; MediaPipe.js in browser
Phase 5 — Hardware	✅ Fully possible	ESP8266 + PCA9685 is a proven, well-documented combo; thousands of working references

The one real constraint acknowledged

MG996R servos are open-loop — no position encoder, no feedback.

- Cannot verify if arm actually reached commanded position
- Under gravity load at full extension → droop
- **This is a design constraint to work around, not a blocker:**
 - Always hold PWM active (never idle the signal)
 - Keep workspace within 15 cm of base
 - Avoid full horizontal extension

Conclusion from assessment

"The project is not just possible — your hardware is overqualified for it."

RTX 3060 alone is significantly more powerful than what most university robotics projects run on. Every software phase has direct ROS2 package support. Every hardware component is a proven, documented design. No blockers exist.

5. Updated Procurement Status (Post This Session)

Item	Previous Status	New Status
DC Jack Female (screw terminal)	Being bought	✅ Ordered
MPU6050	To order	✅ Ordered
400-pt breadboard	To order	✅ Ordered
40-pin jumper wires	To order	✅ Ordered (larger set)
ESP32-CAM + base board	Not in plan	✅ Bought (reserved, not active)

6. MoveIt2 Status Update (From Terminal History)

From terminal history observed in this session:

- `mkdir robotic_arm_moveit_config` — config package directory created
- `ros2 launch robotic_arm_moveit_config demo.launch.py` — launched successfully (exit code 0)

Implication: MoveIt2 is installed and `demo.launch.py` runs. Phase 1 is in progress.

7. Next Steps (Unchanged)

1. Complete MoveIt2 Setup Assistant configuration (planning groups, SRDF, kinematics solver)
2. Verify interactive marker → Plan → Execute works in RViz2 + Gazebo
3. Begin Phase 2 landmark-to-joint mapping ([gesture_node.py](#))
4. When hardware arrives: wire PCA9685 + ESP8266 per wiring diagram

Session ended. Hardware procurement finalised. Feasibility confirmed. No code changes.